🎯 Lean Delivery & Value Stream Mapping

Week 7 • Day 4 • Agile Delivery & ADO Mastery

TPM Academy

Week 7 builds Agile delivery and Azure DevOps (ADO) mastery.

🎯 Day 4 Objectives

- Identify the **value stream** from idea to value-realized (7–10 steps)
- Measure or estimate **process time**, **lead time**, and **flow efficiency** per step
- Find the **top 3 queues** with named cause + candidate intervention
- Draw the **visual VSM**
- Use AI to surface **non-obvious queues**

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin DP Sections 1–3
09:15 – 10:30	Block 1 + Activity 1	Identify your value stream
10:45 – 12:00	Block 2 + Activity 2	Measure or estimate each step
13:00 – 14:15	Block 3 + Activity 3	Top 3 queues
14:30 – 16:00	Block 4 + Activity 4 + Wrap	VSM + AI critique

Block 1 — The Lean Claim

Most of the time, work is waiting

📚 Lean's Foundational Claim

Most of the time work spends in a system is **waiting**, not being worked on. Reduce waiting; reduce total time.

Typical software example: a story with 2 hours of actual coding may spend 3 weeks in the system — waiting for spec, review, QA, deploy slot, merge.

A TPM who optimizes coding time saves minutes. A TPM who reduces the waiting saves weeks.

📚 VSM Vocabulary

Term	Meaning
Value stream	Whole flow from idea to value-realized
Process time (PT)	Hands-on-work time at a step
Lead time (LT)	Total time at a step including waiting
Flow efficiency	PT / LT × 100%
Queue	Work waiting between steps
Cycle time	Lead time across the whole stream
Throughput	Completed items per unit time
WIP	Items currently being worked
Little's Law	Cycle time = WIP / throughput

📚 Typical Software Value Stream

```
Idea → Discovery → Spec → Build → Ship → Measure → Iterate
|       |           |      |       |      |          |
▼       ▼           ▼      ▼       ▼      ▼          ▼
Q1      Q2          Q3     Q4      Q5     Q6         Q7
```

Worst queues for most teams:

- Spec → Build (waiting for engineering capacity)
- Build → Ship (review, QA, deploy slot)
- Ship → Measure (data accumulation; nobody looks)

👥 Activity 1 — Identify Your Stream

Quads • 45 minutes

- 25 min: walk steps idea → value-realized
- 10 min: identify queues between steps
- 10 min: mark step boundaries explicitly

⏱️ 25 + 10 + 10

Block 2 — Measure or Estimate

Process time / lead time / flow efficiency

📊 Worked Numbers (FieldPulse)

Step	PT	LT	Flow eff
Idea	1 hr	3 days	5%
Discovery	4 hrs	2 days	25%
Spec	16 hrs	2 weeks	22%
Estimate + accept	1 hr	1 week	3%
Build	8 hrs/story	1 week/story	14%
Code review	30 min	1.5 days	4%
QA + integration	4 hrs	2 days	12%
Deploy	30 min	6 hrs	8%

Total cycle time: 6–8 weeks. **Total flow efficiency:** 5–15% typical.

👥 Activity 2 — Measure / Estimate

Quads • 50 minutes

- 50 min: PT + LT + flow eff per step

⏱️ 50 min straight

Block 3 — Top 3 Queues

Find where the time hides

🔎 Queue Investigation

For each top queue:

- **Why does this queue exist?** (capacity, policy, inertia, hidden dependency)
- **What would shrink it?** (capacity, policy change, parallel work, dependency removal)
- **Cost of shrinking?** (people-time, money, organizational change)

> **"We're slow in code review"** is not specific. **"PRs wait 2 days for first review because we have 8 reviewers and 40 PRs/week"** is.

📝 Worked Top 3 Queues

Queue	Cause	Intervention candidate
Spec → Estimate (1 wk)	Sprint planning bi-weekly	Mid-cycle sync; mid-sprint pull
Build → Code review (1.5 d)	5 reviewers; PRs wait	Pair-review during dev; auto-assign
Discovery → Spec (varies)	Customer interview scheduling	Async cycles; lighter-weight discovery for known patterns

👥 Activity 3 — Top 3 Queues

Quads • 50 minutes

- 5 min: sum total queue time
- 10 min: identify top 3 queues
- 25 min: investigate each (cause / intervention / cost)
- 10 min: rank by impact / effort / risk

⏱️ 5 + 10 + 25 + 10

Block 4 — Visual VSM + AI

One page; the queues hit emotionally

📝 The VSM Canvas

```
[Step 1]──→[Step 2]──→[Step 3]──→...
|           |           |
PT: __      PT: __      PT: __
LT: __      LT: __      LT: __

QUEUE       QUEUE       QUEUE
__ items    __ items    __ items
__ days     __ days     __ days

═════════════════════════════════════
Total cycle time: __ days
Total process time: __ days
Total flow efficiency: __%
```

🤖 AI Critique Prompt

```
Role: Lean coach reviewing a software value stream.
Context: <description of the VSM with PT/LT/flow eff numbers
and identified queues>
Task: Identify 3 queues likely missing or under-counted:
1. A queue between two steps not yet shown
2. A step where PT and LT seem out of line with team reality
3. An invisible queue (e.g., approvals, "ping someone")
Constraints:
- Use only the data given
- Be specific
Format: 3 numbered findings — what's missing / why suspect /
how to verify.
```

👥 Activity 4 — VSM + AI

Quads • 55 minutes • Wrap

- 30 min: draw the VSM
- 10 min: AI critique
- 10 min: update VSM; recompute flow efficiency
- 5 min: document in Section 4

⏱️ 30 + 10 + 10 + 5 · 15-min wrap

🎉 Day 4 Wrap

What we built

- Value stream identified
- PT / LT / flow efficiency per step
- Top 3 queues with cause + intervention
- Visual VSM
- Delivery Plan (DP) Section 4 drafted

What opens tomorrow

- **Bottleneck removal experiments**
- Hypothesis / test / success criterion
- Friday: full DP integration + sign-off

Bring to Day 5: DP Sections 1–4 + your top 3 queues. Tomorrow we design the experiments.

💬 Questions & Reflection

Where does your team's time hide that nobody talks about?